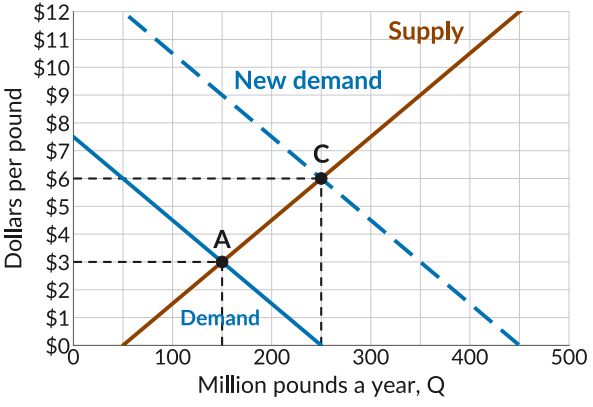
Oil Booms and Busts:
Market Dynamics in the Short Run and Long Run

ECON 201: Principles of Microeconomics
Lecture 19

Div Bhagia

A Shift in Demand

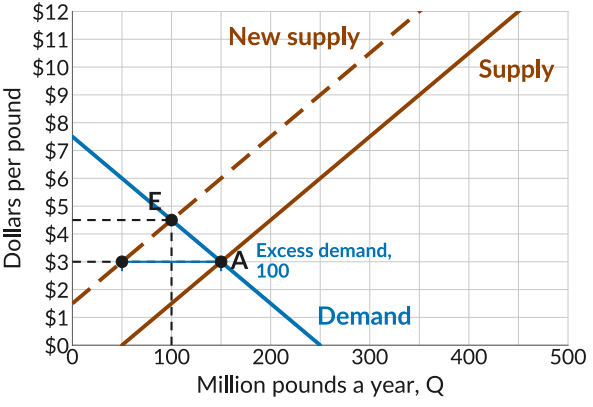
- Last class, demand for quinoa rose at every price: a **shift in the demand curve**.
- At the old price of $3 there was **excess demand**, so the price rose and farmers grew more: a **movement along the supply curve**.
- The market moved from A to C: from 150 million pounds at $3 to 250 million pounds at $6.



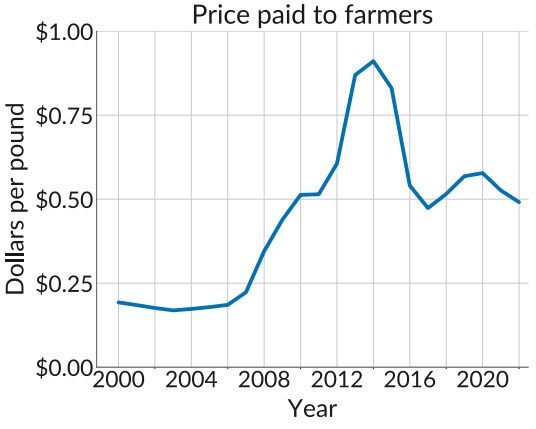
A Shift in Supply

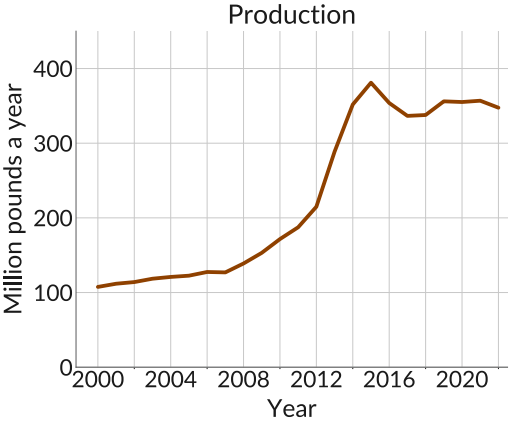
The supply curve can shift too. As a hypothetical, suppose that before the boom a drought raised the cost of every pound by $3.

- Supply shifts up by $3, the same as left by 100.
- At $3, farmers grow 50 but buyers want 150: an **excess demand** of 100.
- The price rises until the market clears at E, 100 million pounds at $4.50: a movement along the demand curve.



The Quinoa Boom

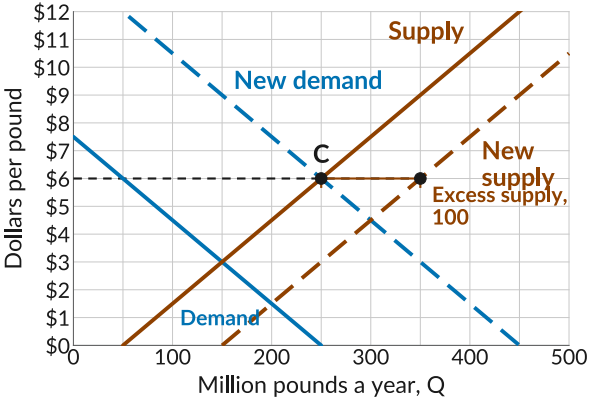




A shift in demand alone would have kept the price high. So why did it come back down?

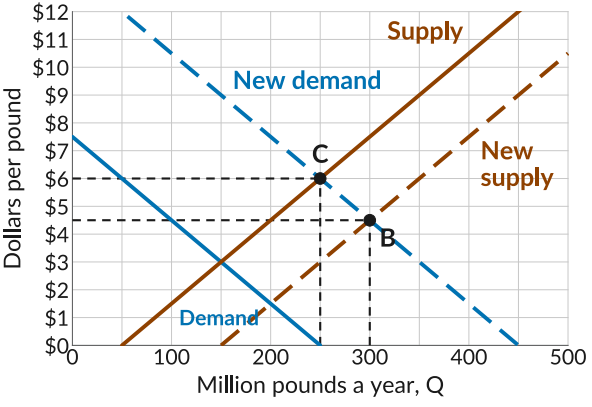
More Land, More Farmers

- In the **long run**, high prices drew more land and more farmers into quinoa, so a **shift in the supply curve**.
- At $6, farmers would grow 350 million pounds, but buyers want 250: an **excess supply** of 100.
- Growers with unsold quinoa accept lower offers, and farmers whose costs exceed the lower price grow less, until the market clears.



The Price Comes Back Down

- The market clears where the new demand curve meets the new supply curve, at B: 300 million pounds at $4.50.
- Price falls and quantity rises. Demand did not shift: the quantity demanded rose from 250 to 300 because the price fell, a movement along the demand curve.
- This explains why quinoa prices stabilized again.



Why Does the Supply Curve Shift?

Short run: the number of firms and their capacity are fixed.

- A new technology lowers marginal cost, so supply shifts right.
- A drought raises marginal cost, so supply shifts left.
- Producers agree to sell less, or a war disrupts production, so supply shifts left.

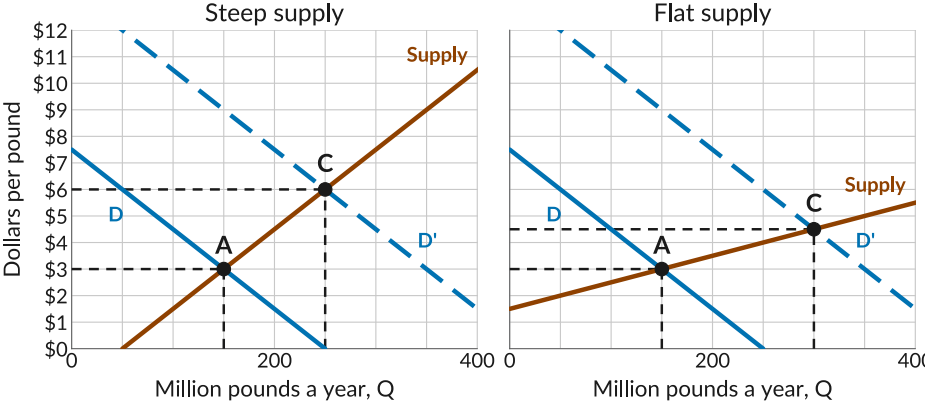
Long run: firms can invest, enter, or leave.

- Economic profit draws in new firms and new capacity, so supply shifts right.
- Losses drive firms out of the market, so supply shifts left.

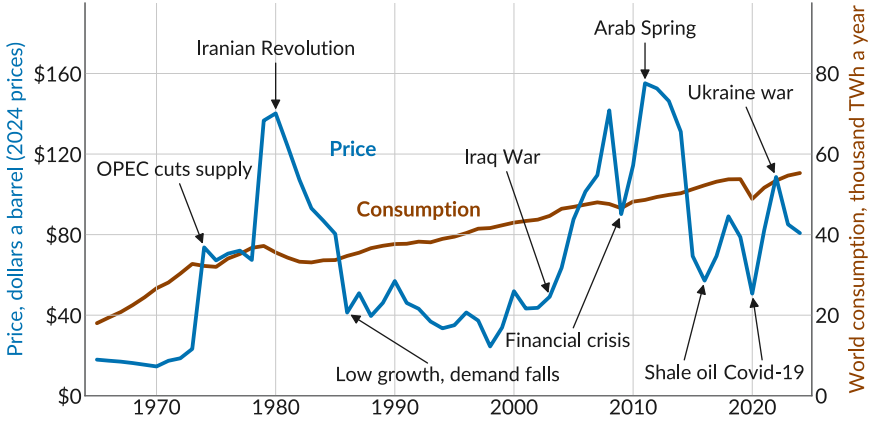
Another way of thinking of the long run: supply is more elastic.

When Supply Is Steep

The same surge in demand, with a steep and a flat supply curve through A.



The Oil Market



Source: Our World in Data (Energy Institute); prices adjusted for inflation with the US consumer price index.

Why Did the Oil Price Move?

- **1970s, supply shifts left**: OPEC, a cartel, restricted its supply, and the Iranian Revolution cut output.
- **1980s and 1990s, low prices**: low growth shifted demand left. In the long run, the high prices of the 1970s also lowered demand, as users cut back, and raised supply, as new producers entered.
- **2000s, supply shifts left**: the Iraq War and later the Arab Spring limited supply, while the financial crisis cut demand for a short time.
- **Since 2014, mixed shifts**: shale oil shifts supply right, Covid-19 shifts demand left, and the Ukraine war shifts supply left.

Is Competitive Equilibrium a Good Model?

Competitive equilibrium needs four conditions: (i) many buyers and sellers acting independently, (ii) identical goods, (iii) everyone aware of the prices others trade at, and (iv) everyone seeking the best price.

- When they hold, every trade happens at one price, the **Law of One Price**, and the market is called **perfectly competitive**.
- They rarely hold exactly. Oil had too few large sellers to stop a cartel; many goods are differentiated; buyers do not always search for the lowest price.
- Two gas stations on the same road can charge different prices, because checking prices takes time and many drivers stop at the first one they see: conditions (iii) and (iv) fail.

Where the Model Works

- Competitive equilibrium is a useful **benchmark**: it gives the largest total surplus the market can produce.
- It fits markets for farm goods like wheat, rice, coffee, and quinoa, with many sellers of nearly identical products.
- It is a fair approximation for everyday goods sold by many similar shops, like a loaf of bread or a cup of coffee.

Practice: Lecture 18's practice problems cover this lecture too.